PROJECT PRESENTATION

MERN-Based Exam Preparation Dashboard

A web application to track study progress, manage subjects, and monitor test performance.

PLAN • PRACTICE • IMPROVE



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01 | Problem Statement & Objectives

Why a centralized exam-preparation dashboard is useful

THE PROBLEM

- Study plans are scattered across notebooks and apps.
- Students may lose track of completed and pending topics.
- Marks alone may not reveal weak subjects or progress trends.
- Without a history, it is hard to plan revision effectively.

PROJECT OBJECTIVES

- Organize subjects, topics, and deadlines.
- Record study sessions and topic completion.
- Store test scores and show performance trends.
- Identify weak areas and plan focused revision.
- Provide a simple, responsive user experience.

02 | Core Features

Key workflows for students

01

Study Planner

Set subjects, topics, deadlines, and daily goals.

02

Progress Tracking

Mark topics as pending, in progress, or completed.

03

Test Score Tracker

Record marks, dates, and subject-wise results.

04

Analytics Dashboard

See completion rates, averages, and score trends.

05

Revision Support

Flag weak topics and schedule extra practice.

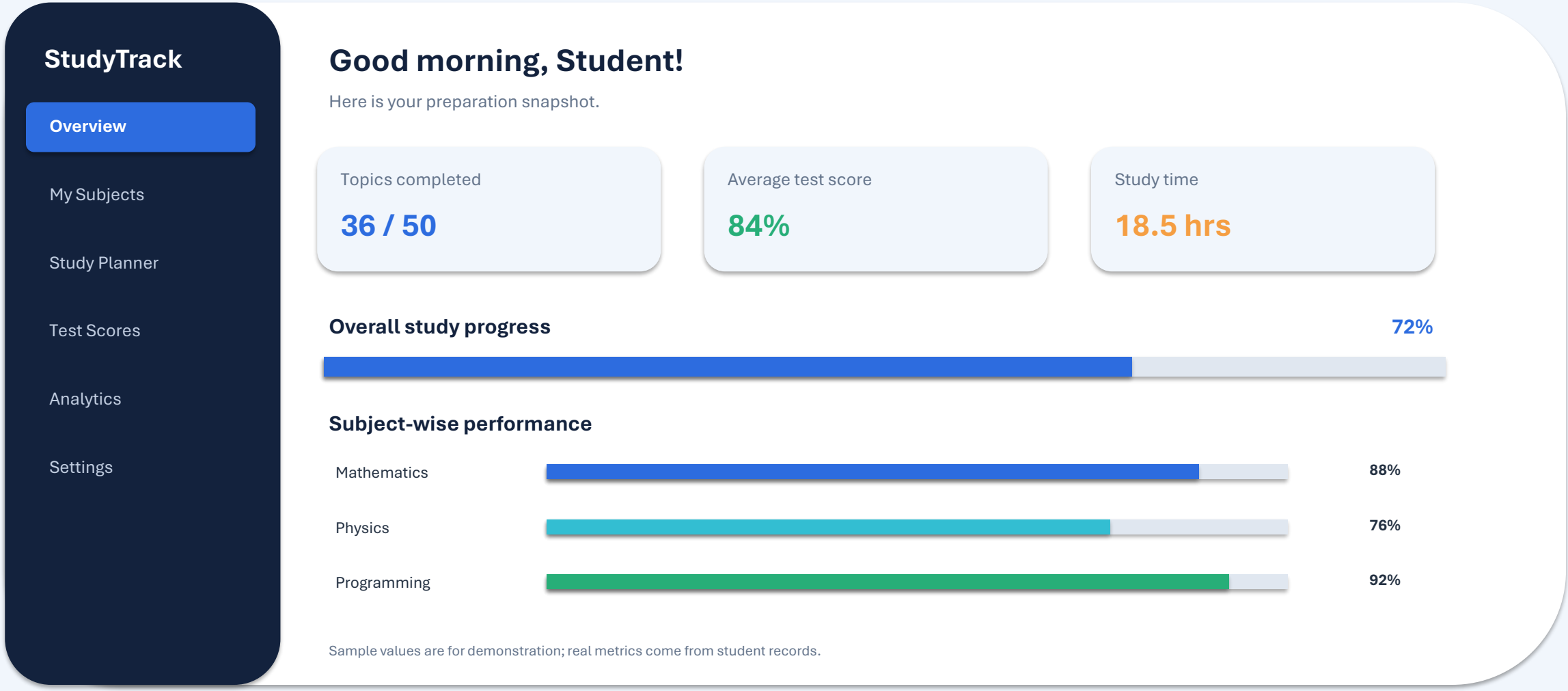
06

User Accounts

Keep each student's information separate.

03 | Dashboard Design

Illustrative dashboard using sample values



04 | MERN Stack Architecture

How the frontend, backend, and database work together

FRONTEND

React.js

Screens, forms,
charts and navigation

→

API LAYER

HTTP / JSON

Requests and responses
through REST APIs

→

BACKEND

Node + Express

Business logic, validation
and authentication

→

DATABASE

MongoDB

Users, topics, sessions
and test scores

TYPICAL DATA FLOW

Student action → React sends request → Express validates and processes → MongoDB saves or returns data → React refreshes the dashboard.

Security: hash passwords, protect routes, validate inputs, use environment variables, and enforce user-specific access.

05 | Data Model & API Design

A practical starting structure for the backend

DATABASE COLLECTIONS

User	name, email, passwordHash
Subject	userId, title, examDate
Topic	subjectId, title, status, targetDate
StudySession	userId, topicId, duration, date
TestResult	userId, subjectId, score, totalMarks, testDate

IDs connect records across collections.

EXAMPLE REST ENDPOINTS

- POST /api/auth/register
- POST /api/auth/login
- GET /api/subjects
- POST /api/subjects
- PATCH /api/topics/:id
- POST /api/tests
- GET /api/analytics/summary

Middleware should ensure students access only their own records.

06 | Tracking Logic & Analytics

Simple calculations create useful feedback

TEST PERCENTAGE

(Score ÷ Total) × 100

42 / 50 = 84%

TOPIC COMPLETION

(Done ÷ Total) × 100

36 / 50 = 72%

AVERAGE TEST SCORE

Sum of percentages ÷ tests

Compare scores over time

STUDENT FEEDBACK LOOP

1. **Plan topics** →
2. **Log study time** →
3. **Take a test** →
4. **Review weak areas** →
5. **Update plan**

Use trends to guide revision; do not judge preparation from one test alone.

07 | Technology & Implementation Plan

Recommended tools and a phased build process

TECHNOLOGY STACK

React.js	Reusable interface components
Node.js	JavaScript runtime
Express.js	REST APIs and middleware
MongoDB	Document database
Mongoose	Schemas and database queries
Chart.js / Recharts	Progress and score charts

BUILD IN PHASES

Phase 1
Set up project, database, routes, and login.

Phase 2
Build subjects, topics, study logs, and test entry.

Phase 3
Add metrics, charts, and analytics filters.

Phase 4
Test validation, mobile layout, and security.

08 | Testing & Expected Outcomes

Check correctness, usability, and safe data access

TEST CASES

Progress	Completed topic updates percentage.
Score	42 out of 50 displays as 84%.
Validation	Reject missing, negative, or impossible marks.
Access	Users cannot read each other's records.
Responsive UI	Works on desktop and mobile screens.

EXPECTED BENEFITS

- Clear view of pending and completed topics.
- Quick access to test history and trends.
- Focused revision based on weaker subjects.
- Better planning through measurable goals.
- One organized space for exam preparation.

Results improve with regular and accurate data entry.

Make preparation measurable.

This MERN application brings planning, progress tracking, and test analytics into one responsive dashboard.

FUTURE ENHANCEMENTS

- AI-assisted study plans
- Reminders and calendar integration
- Question banks and mock exams
- Exportable reports and streaks

FINAL TAKEAWAY

Track what you study. Measure what you learn. Improve what comes next.

THANK YOU • QUESTIONS?